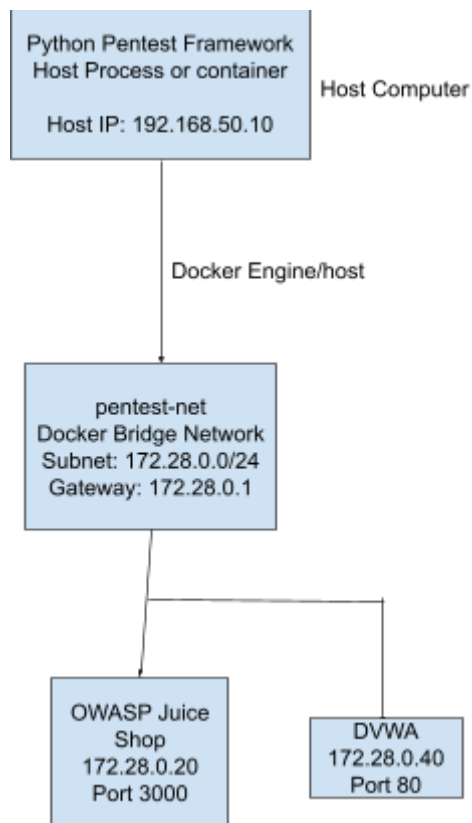


The network architecture I plan to implement will provide a controlled and secure Docker-based environment for developing and testing the framework. The environment will contain the Python penetration-testing framework and intentionally vulnerable web applications such as OWASP Juice Shop and DVWA.

Instead of using separate physical devices, the proposed implementation will use Docker containers connected through a dedicated user-defined Docker bridge network. This provides logical network isolation while allowing the containers to communicate with one another using private IP addresses and Docker DNS service names.



Docker environment will use the private IPv4 network

Docker Network: `pentest-net`

Network: 172.28.0.0/24

Subnet Mask: 255.255.255.0

Docker Gateway: 172.28.0.1

Broadcast address: 172.28.0.255

The /24 subnet is enough as the environment will only have a few containers.

OSI Layer	Layer	Example
7	Application	Python Penetration-Testing Framework HTTP / HTTPS Requests
6	Presentation	TLS encryption HTTP data formatting
5	Session	Web sessions Cookies and authentication sessions
4	Transport	TCP connections and ports such as 80, and 3000
3	Network	IPv4 addressing within 172.28.0.0/24
2	Data Link	Docker virtual Ethernet interfaces and bridge networking
1	Physical	Host computer hardware and physical network interface

The penetration-testing framework will primarily interact with the Application, Transport, and Network layers.

TCP/IP Layer	Example in Project
Application	Python / HTTP / HTTPS
Transport	TCP Port 3000 or Port 80
Internet	IPv4 172.28.0.10 → 172.28.0.20
Network Access	Docker bridge and virtual Ethernet

Proposed Security Protocol

Protocol or control	Purpose
HTTPS	Secure web communication
TLS	Encrypts application traffic
TCP/IP	Network communication between containers

Docker DNS	Resolves service names within the Docker network
SSH	Secure remote administration of the host
Host firewall	Restricts access to published container ports
Docker network isolation	Separates vulnerable applications from the normal LAN